An Advanced Data Security Algorithm Using cryptography and DNA-Based steganography

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***Abstract*—nowadays, data security has become one of the most important branches of computer sciences because of the great development in the Internet field and the great need to exchange information over the Internet. Cryptography is to protect the content of messages. The main problem in cryptography is that the data is not hidden. Although the data cannot be read, but it still exists so, someone can decrypt it. Steganography is about concealing the data existence. In this paper a new approach is introduced to ensure the secrecy and integrity of data transported over computer networks. The paper describes how cryptography and steganography were combined to produce higher level of security. Advanced Encryption Standard (AES) is used to encrypt data and a steganography technique is used to insert encrypted data in the deoxyribonucleic acid (DNA) sequence and then apply the central dogma of the molecular biology processes which converts DNA form to ribonucleic Acid (RNA) form then to protein form. The proposed technique was applied successfully.**

Keywords—cryptography; steganography; DNA-based steganography; Transcription; Translation.

I. INTRODUCTION

The security is an active area of research, especially after the tremendous development in the communication field and the Internet, it aims at keeping information safe by using cryptography and/or steganography, steganography is not to be confused with cryptography. Cryptography is a group of techniques used to convert information into a non-readable form through the use of an encryption key and an encryption algorithm. Steganography on the other hand hides the message inside another medium (e.g. an image, audio file, DNA…etc.). Cryptography is to protect the content of messages [1]. The main problem in cryptography is that the data is not hidden and could be compromised to get the original message. Steganography is about concealing the data existence [1]. DNA-based cryptography and steganography is a modern technique born from Adleman's research [2] in DNA computing. The research in the DNA field is very important because of its properties that can be utilized in many fields. DNA has an amazing density as an information storage medium, that a single cubic centimeter of DNA holds more information than a trillion CDs. A gram of DNA contains 10^{21} DNA bases which can correspond to 108 terabytes of binary data [3].

The paper starts with a brief introduction to the principles of molecular biology. Related work is then described. Following that, the proposed algorithm is explained in details, from data preparation to encryption using AES and finally DNA-based steganography through injecting the encrypted data into a real DNA sequence. The central dogma of molecular biology is then applied to the resulting DNA sequence to create the corresponding protein. As part of the research, an implementation of the algorithm was created and tested. Test results indicated that the proposed algorithm offers considerable performance improvements compared to its predecessors. The test results are detailed in the paper and performance comparison is also illustrated.

II. OVERVIEW ON MOLECULAR BIOLOGY

It is important to know some information about the synthesis of DNA, DNA properties and the processes that occur on DNA in order to understand the proposed algorithm.

A. DNA

Deoxyribonucleic acid (DNA) molecule consists of two long polynucleotide chains composed of four types of nucleotide subunits [5]. DNA is built up of a sugar-phosphate backbone and a sequence of nucleotides: adenine (A), guanine (G), cytosine (C) and thymine (T). Each base can only pair up with one single predetermined other base: A+T, T+A, C+G, G+C as illustrated in the figure1.

Each DNA molecule is composed of two intertwined strands of single DNA chains, bonded together by hydrogen bonds: A = T double bond and C ≡ G triple bond [6].

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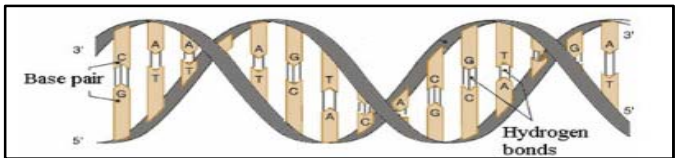

Figure 1. DNA structure [6]

B. RNA

Ribonucleic Acid (RNA) is a nucleic acid polymer consisting of nucleotide bounds, It always a single stranded molecule. The backbone is formed of a slightly different sugar from that of DNA ribose instead of deoxyribose [7]. RNA is built up of four bases: adenine (A), guanine (G), cytosine (C) and uracil (U). The first three (A, C, and G) are the same nucleotide bounds in DNA but uracil instead thymine. DNA converts to RNA by enzymes in the transcription process.

C. Protein

A protein is a linear polymer of amino acids linked together by peptide bonds. The amino acids are quite different from those of DNA and RNA, and there are 20 types, instead of 4 [7]. Each amino acid has a three-letter code.

D. central dogma process

The central dogma is the process that converts DNA to RNA by transcription process and then converts RNA to Protein molecules by translation process, in which the amino acid sequence is determined by the nucleotide sequence of the DNA as shown in figure 2.

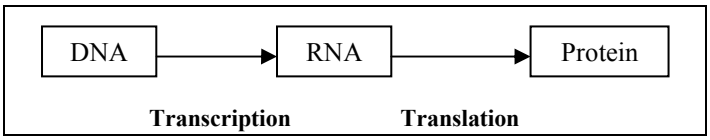

Figure 2. Central dogma process

The mRNA is a type of RNA for making protein. Messenger RNA is mainly in charge of the protein synthesis in the cell which is manufactured by the ribosome. The RNA transcript is referred to as messenger RNA (mRNA) [8]. The information in the messenger RNA molecule sequence is read out in groups of three nucleotides at a time: each triplet of nucleotides called codon will translate to a single amino acid in a corresponding protein.

III. RELATED WORK

Several previous researches were studied. The most relevant of them was Tantawi's model [4] that employs cryptography and steganography in DNA using the following stages:

First, steganography; this is done in two steps, Data Formatting and Data Hiding. Data formatting method works by taking "plain text" in text format as an input and encoding it to DNA strands. After that Data Hiding is performed: in the steganography process, a DNA key structure is used with XOR operation to hide the DNA form of the message.

Second, cryptography; this is also done in two steps DES Encryption Process and The Biological Process Analogy. DES Encryption is used to encrypt the steganography -DNA from of Data. After that The Biological Process Analogy is applied in a similar fashion to the DNA biological transcription, splicing and translation process consecutively.

IV. THE PROPOSED ALGORITHM

The **c**ryptography and **D**NA-based **s**teganography algorithm − hereafter referred to **CDS** algorithm − is used to hide a picture or text in the real DNA sequence by applying the following encoding stages:

1. *Encryption phase:*
   The encryption phase is performed into two steps as depicted in figure 3, and is detailed below:

A. Convert the image into a byte array and text into ASCII code.

An image is composed of pixels. Each pixel is a combination of basic colors. Every color value ranges between (0-256). Each value represents a certain color, and this is how the image is formed on the computer and on this basis the image will be converted to an array of bytes, so that the current algorithm can be applied to images. Likewise, the text is a collection of letters every letter is represented by an ASCII code. These letters will be converted to ASCII codes that will be encrypted and hidden inside the DNA sequence.

B. Encrypt data by AES algorithm.

The AES (Advanced Encryption Standard) was designed by Vicent Rijmen and Joan Daemen in Belgium. The AES was adopted to use by the U.S. government in December 2001[9]. AES has 128/192/256 bits key size and 128 bits data block, depends on the key size that will determine the number of rounds 10/12/14. In this paper AES encryption is used with 128-bits key, 128 bits data block and 10 rounds. The Round Key of each round is generated from the cipher key using the Key Expansion algorithm. The AES is designed to be simple and at the same time resistance against known attacks and it is fast. The AES has code compactness on many processors and this property will increase the speed of time processing.

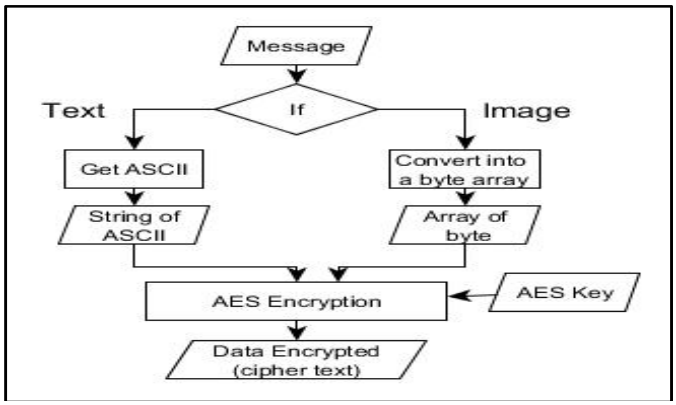

Figure 3. Encryption phase

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2. *steganography phase:*
   This is performed into two steps as depicted in figures 5 and 6 is detailed below:

A. Convert encrypted data to binary (zero, one) form.

The output of the encryption process is a string of characters. These characters are converted into binary form through mapping each character to its corresponding binary ASCII code.

B. Insert data to DNA sequence.

The DNA is a sequence of nucleotides: adenine (A), cytosine (C), guanine (G) and thymine (T). Each nucleotide will take a binary value according to the following table:

TABLE1. Nucleotides to binary value

Bases	Binary value
A	00
C	01
G	10
T	11

These binary values need to be converted from DNA form to binary form and vice versa. This method is suggested by [10]. Data insertion into DNA sequence will be performed through the following steps (also depicted in Fig 5):

1. Convert data (M) into binary format.
2. Convert DNA sequence (S) to binary format using table 1. The DNA sequence is the medium used for hiding messages inside a real human DNA. This DNA sequence is acquired from public available biology database [11].
3. Divide (S) into 1-bit segments.
4. Bit-wise insertion of M into the beginning of each segment of S. If S is longer than M, then the remaining S sequence is truncated.
5. Convert the resulting binary format into DNA format according to table 1.

Example of the data insertion method is illustrated in figure 4.

Assuming:
M: Plain Message
S: DNA Sequence
The following example will help explain the above steps:
Insert (HI) to DNA
M = "HI"
S = "AAGCTAAT"
Step1.
M= **01 00 10 00 01 00 10 01**
Step2.
S = **00 00 10 01 11 00 00 11**
Step3.
S= **0,0,0, 0,1,0, 0,1,1,1,0,0,0,0,1,1**
Step4.
00,10,00,00,11,00,00,01,01,11,00,00,10,00,01,11
Step5.
"AGAATAACCTAAGACT"

Figure 4. Example of data insertion to DNA

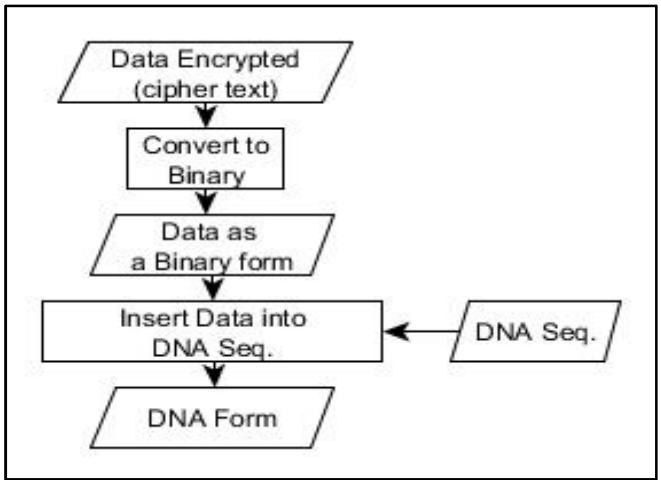

Figure 5. Insertion method

C. Use central dogma process to convert DNA sequence form into protein form.

The central dogma has two processes the first one is the transcription. In this process, DNA is converted to RNA form. The second process is the translation process in which RNA is converted into protein form as illustrated in figure 6. The central dogma processes are described as follows:

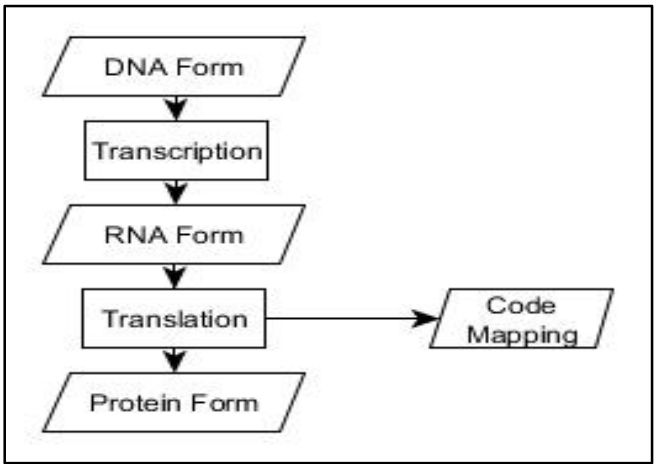

Figure 6. Central Dogma

1. TRANSCRIPTION PROCESS

It is described as follows:

- The DNA complementary strand is generated by converting each 'A' to 'T', 'T' to 'A', 'C' to 'G' and 'G' to 'C' as:

  DNA form: "AGAATAACCTAAGACT"
  DNA complement: "TCTTATTGGATTCTGA"
- converting The DNA complementary strand form into the mRNA form by replacing 'T' by 'U' as:

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DNA complement: "TCTTATTGGATTCTGA"
mRNA form: "UCUUAUUGGAUUCUGA"

2. *Translation Process*

In this process, the mRNA sequences will be divided into groups. Each one consists of three nucleotides called codon such as "UCU, UAU, UGG, AUU, CUG, AAA" and then translates this codon into the corresponding amino acid. mRNA codons are three nucleotides long. There are 4 possible nucleotides A, C, G, and U, which means that there are (4 x 4 x 4) or 64 unique codes that can be generated. But, there are only 20 amino acids that are encoded. That means we have redundancy problem that will be sets of 1,2,3,4 or 6 codons representing the same amino acid. To solve this problem genetic code is used in Table 2 (64 codons to 20 amino acids).

This table will solve the codon redundancy problem by giving numeric value from (0 to 5) to the codon representing the same amino acid. This suggestion by Boris Shimanovsky [8], as follows:

mRNA form: "UCU, UAU, UGG, AUU, CUG, AAA"

Protein form:"Ser, Tyr, Trp, Ile, Leu, Lys"
Value: " 5, 1, 0, 2, 2, 1

Where, the table header letter V: value, C: codon and A: amino acid.

But, there are two problems. First, after dividing the RNA sequence into codons (groups of three nucleotides), sometimes the RNA sequence doesn't accept the division by three; so the result will be similar to:

"UCU, UAU, UGG, AUU, C"
"UCU, UAU, UGG, AUU, CU"

To solve this problem the letter 'A' will be added once or twice at the end of the sequence to complete the last codon like these:

"UCU, UAU, UGG, AUU, CAA"
"UCU, UAU, UGG, AUU, CUA"

Second problem, when receiver decode the message how could he know the letter 'A' if it is from the original message or has been added to sequence in the central dogma process to manipulate codon problem, and to solve the problem send the size of the bits that inserted in the DNA sequence to the receiver. Now the receiver knows the number of bits that must be retrieve from protein form. Code mapping is containing the codon amino acids value to solve the redundancy problem.

It's observed that the resulting protein form has a size that's eight times the size of the original message. This is a potential area of improvement.

TABLE 2. The RNA - Amino Acid Mapping

V	A	C	V	A	C	V	A	C	V	A	C
0	Ala	GCA	0	Gln	CAA	3	Leu	CUU	4	Ser	UCG
1	Ala	GCC	1	Gln	CAG	4	Leu	UUA	5	Ser	UCU
2	Ala	GCG	0	Glu	GAA	5	Leu	UUG	0	(STOP) !!!	UAA
3	Ala	GCU	1	Glu	GAG	0	Lys	AAA	1	(STOP) !!!	UAG
0	Arg	AGA	0	Gly	GGA	1	Lys	AAG	2	(STOP) !!!	UGA
1	Arg	AGG	1	Gly	GGC	0	Met	AUG	0	Thr	ACA
2	Arg	CGA	2	Gly	GGG	0	Phe	UUC	1	Thr	ACC
3	Arg	CGC	3	Gly	GGU	1	Phe	UUU	2	Thr	ACG
4	Arg	CGG	0	His	CAC	0	Pro	CCA	3	Thr	ACU
5	Arg	CGU	1	His	CAU	1	Pro	CCC	0	Trp	UGG
0	Asn	AAC	0	Ile	AUA	2	Pro	CCG	0	Tyr	UAC
1	Asn	AAU	1	Ile	AUC	3	Pro	CCU	1	Tyr	UAU
0	Asp	GAC	2	Ile	AUU	0	Ser	AGC	0	Val	GUA
1	Asp	GAU	0	Leu	CUA	1	Ser	AGU	1	Val	GUC
0	Cys	UGC	1	Leu	CUC	2	Ser	UCA	2	Val	GUG
1	Cys	UGU	2	Leu	CUG	3	Ser	UCC	3	Val	GUU

To decode a message (text, image) from the protein form, the same stages will be applied but in reverse as in the following steps:

1. *steganography phase:*
   A. Use central dogma process in reverse order to retrieve the DNA sequence from the protein sequence.
   B. Retrieve data from the DNA sequence.
   C. Convert data from binary form (zeros and ones) to string (cipher text).

2. *Encryption phase:*
   A. Decrypt data using AES algorithm.
   B. Convert data from byte array into image or ASCII code into text.

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V. EXPERIMENT RESULT

This paper uses C# programming language to implement the proposed algorithm under Visual Studio 2010. The Program can be executed under Windows without any adjustment. It has a windows interface. In this program, the user selects a file type (Image or Message) as shown in figure7.

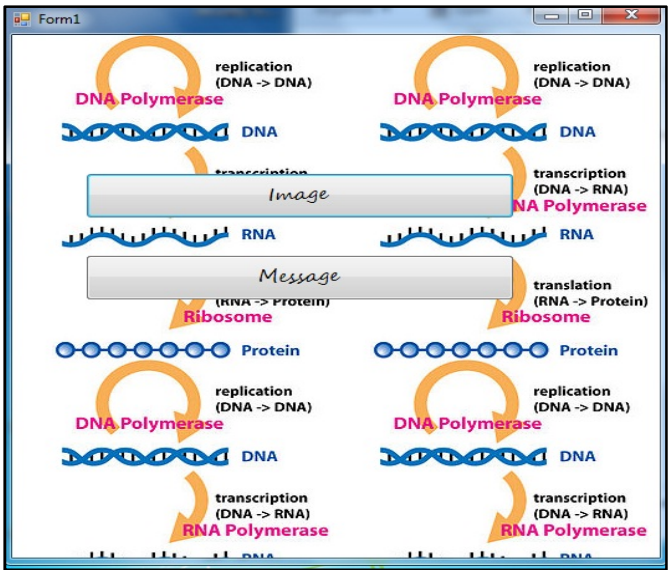

Figure 7. GUI for the proposed application prototype

After select (Image or message) the program will open another form. Then enter the AES key (If it is larger than 16 characters, additional text will be truncated, if it is less, a text will be repeated) after that, the user selects the DNA sequence and then execute as shown in figure 8. The result will be a protein form as a text file and another text file containing the mapping code (codon value), size of the data that has been inserted in the DNA sequence and AES key. These files will be sent to the receiver.

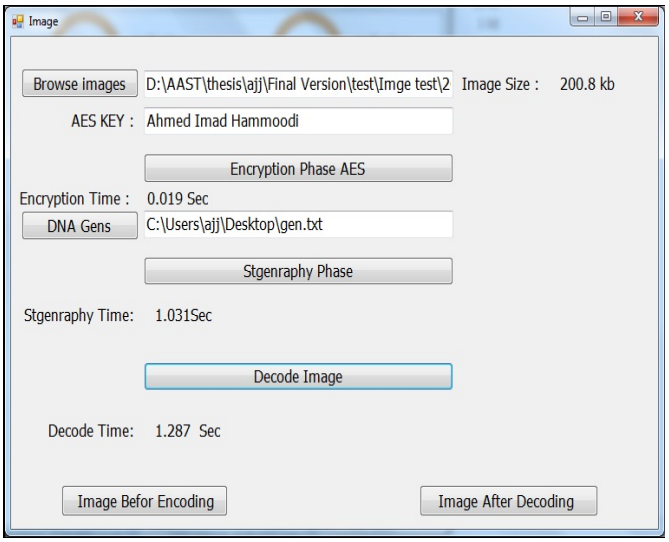

Figure 8. Image form

The experiments were performed using random data on a PC with 2.00 GHz CPU and 1.00 GB RAM under windows 7 32-bits operating system. Different data lengths have been used. For each entry, the experiment was performed 5 times and the average value entered in table 3. The exact same environment was used to produce results data for both the Tantawi's and the CDS algorithms for performance analysis purposes.

TABLE3: processing time for CDS algorithm and pervious work

Test NO.	DATA Size (bytes)	CDS algorithm		Tantawi algorithm	
		Encoding time (ms)	Decoding time (ms)	Encoding time (ms)	Decoding time (ms)
1	1.000	8.41	10.75	22.86	22.91
2	2.000	16.09	18.14	45.72	45.82
3	3.000	23.60	24.72	68.58	68.73
4	4.000	27.96	29.18	91.44	91.64
5	5.000	31.29	33.90	114.30	117.89
6	6.000	34.88	36.39	137.16	141.47
7	7,000	37.74	39.24	160.02	165.05
8	8,000	39.26	43.12	362.58	362.58
9	9,000	43.61	48.82	407.91	407.91
10	10,000	48.96	55.25	453.23	465.22
11	11.000	56.57	62.62	498.55	511.74
12	12.000	64.14	66.87	543.87	558.26
13	13.000	72.16	76.79	560.82	560.82
14	14.000	77.24	82.02	603.96	603.96
15	15.000	82.15	88.25	647.10	658.30

The CDS algorithm offers significant performance enhancements with linear growth with respect to input message size compared to non-linear growth in previous algorithms which exhibited significant performance deterioration with larger files. Figure 9 depicts the performance characteristics of both algorithms for various message sizes.

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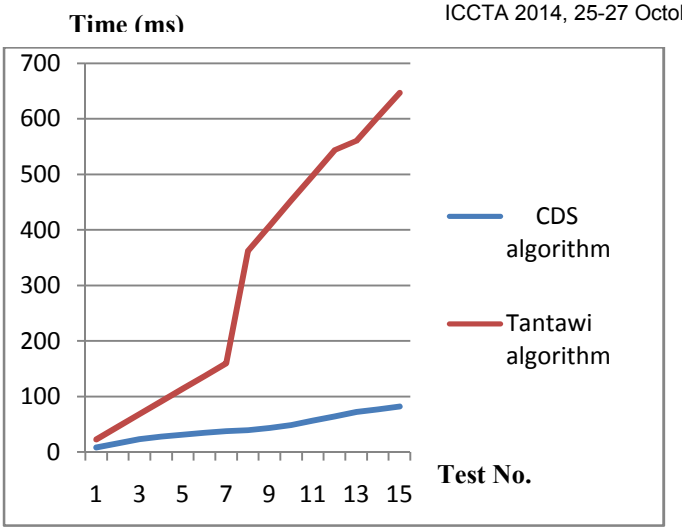

Figure 9. Performance characteristics of both algorithms for various message sizes

The algorithm is applicable to both text and image files of any size, as opposed to the previous algorithms. Testing was performed on both text and image files of various sizes to assess the required processing time in each case. The findings are listed in table 4 and are plotted in figure 10.

TABLE4. Processing time for images and text files of various sizes

Test No.	DATA Size in Kb	CDS algorithm On Image		CDS algorithm On Text	
		Encoding time (ms)	Decoding time (ms)	Encoding time (ms)	Decoding time (ms)
1	50	308.01	354.99	309.75	359.05
2	100	620.61	709.60	620.49	710.45
3	150	912.93	1076.78	911.32	1074.53
4	200	1236.98	1458.08	1234.12	1464.79
5	250	1565.16	1856.67	1558.25	1845.66

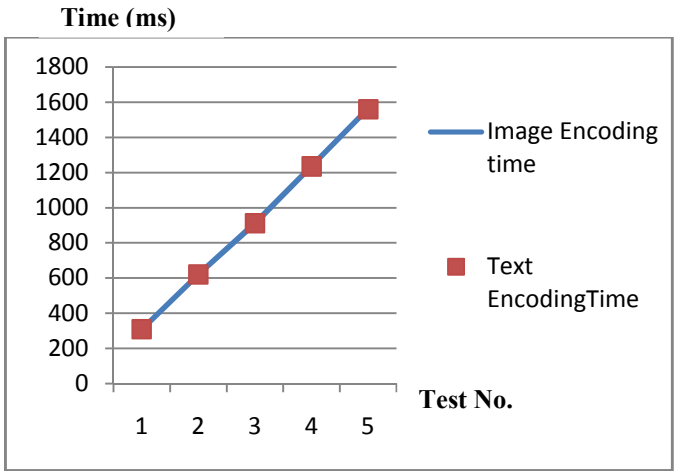

Figure 10. Performance characteristics for images and text file

From figure (10) we conclude that there is practically no difference in the processing time between text file and image files of the same size, as a result the two curves are almost identical. Because the same idea will applied on text and image, the text will be convert to ASCII and then to binary (zero/one), on the other hand the image will convert to array of byte and then to binary.

To verify data integrity, the CDS algorithm was applied to several sample files, and then the reverse process will perform to retrieve the original message. Each retrieved message is compared to corresponding original message through the MD5 algorithm [12]. For all samples, the integrity was maintained.

The distinction between the CDS algorithm and previous work is an important thing, and is summarized in table 5.

TABLE5. CDS algorithm vs. previous work

Factor	Tantawi algorithm	CDS (The proposed algorithm)
Approach	Converts data to DNA form without using real DNA sequence.	Uses real DNA sequence to make steganography. Thus, the encoded message will be more realistic.
Data Format	Text only.	Image and Text.
Data Size	In KB, not good for large size of data.	In MB, good for large size of data.
Encryption Technique	DES (vulnerable)	AES (more secure and difficult to break)
Steganogr-aphy Technique	Using steganography with XOR method and DNA key to convert data to DNA form. While this method is related to cryptography techniques.	Using steganography with insertion method to insert data to DNA sequence.
Central Dogma Implement-ation	Splicing, Transcription and Translation.	Transcription and Translation only.
Splicing Effect	Splicing increases the complexity and make cryptanalysis harder, but it causes major performance deterioration	Splicing was removed, nevertheless achieving the same level of complexity without sacrificing performance

VI. CONCLUSION

The CDS algorithm keeps information secure by using cryptography and DNA-based steganography. Cryptography ensures the security of data communication over an insecure medium in order to provide confidentiality and integrity. On the other hand steganography hides the existence of a message in a real DNA sequence. This paper use DNA properties and a simulation of the processes that occur naturally on the DNA to increase security of information as well as combining cryptography and DNA steganography techniques together to achieve a higher level of security. The CDS algorithm offers

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significant performance enhancements (compared to previous work) with linear growth with respect to input message and it's applicable to both text and image messages. One possible improvement is to reduce the size of the resulting blinded message WRT to the original message as it has 8:1 ratio at the moment.

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